



Review article

Smoking Prevention and Cessation Programs for Children and Adolescents Focusing on Parental Involvement: A Systematic Review and Meta-Analysis



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A B S T R A C T

This study aims to determine the effect and equity outcomes of smoking prevention or smoking cessation interventions for children and adolescents involving parents. A systematic literature search was conducted between 24 November 2022 and 27 November 2023 in PubMed, Medline, Web of Science, Embase, PsycINFO, Google Scholar, ClinicalTrials.gov, EU Clinical Trials Register, and the WHO international clinical trials registry. Experimental or quasi-experimental studies reporting smoking initiation among never smokers, smoking cessation among smokers, and differential effects in socio-economic subgroups with a follow-up of at least 6 months were selected. Information was extracted concerning setting, study design, sample size, type of parental intervention, follow-up time, and relative risk estimates of intervention effects (risk ratios, odds ratios, and hazard ratios). Random effect model was used for meta-analysis of 24 studies, with the remaining studies included in a narrative synthesis. After screening 6,748 records, 36 studies, based on 29 unique interventions (27 smoking prevention and 2 smoking cessation), were included. The summary estimate of relative risk of smoking initiation among children and adolescents participating in smoking prevention interventions involving parents compared to control conditions was 0.84 (95% CI: 0.76, 0.94). Results concerning smoking cessation and equity aspects were inconclusive due to the paucity of studies. Smoking prevention interventions for children and adolescents where parents are actively involved appear to be effective, but the overall quality of evidence was moderate. No conclusion was possible to be drawn on the effects of interventions involving parents on offspring's smoking cessation. Equity aspects of these interventions remain to be studied.

IMPLICATIONS AND CONTRIBUTION

This meta-analysis demonstrates that smoking prevention interventions with parental active involvements are moderately effective to delay the onset of smoking among children and adolescents. This study contains updated evidence and brings some important aspects forward, e.g., assessing the equity aspects of smoking prevention and cessation interventions in future.

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Tobacco use is one of the leading causes of global morbidity and mortality [1]. Annually, more than 8 million deaths worldwide can be attributed to tobacco use, of which more than 7 million result from first-hand tobacco use [2]. The risk of

initiating a smoking behaviour is highest among adolescents [3]. A 40-year retrospective cohort study among the European population showed that the rate of smoking initiation was highest among 15–16-year-old adolescents in the 2000s [4]. According to Health Behaviour in School-aged Children survey 2017–2018, based on adolescents from 45 countries in Europe and North America, the lifetime prevalence of tobacco use was 29% and 27% among 11–15-year-old boys and girls, respectively [5].

Conflict of Interest: The authors declare no competing interests.

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As childhood and adolescence are critical periods to initiate smoking, targeting these groups in the smoking prevention and cessation interventions is a priority, given that early age is a strong predictor of escalation and maintenance of smoking in adulthood [6–8].

Determinants of smoking among children and adolescents are to be traced in the behavioral, cognitive, emotional, and environmental domains [7,9,10]. Among the factors strongly predicting smoking initiation, parental influence is the most important at least up to late adolescence [7,11–16]. This pathway is also supported by behavioral theories such as ecological system theory [17]. Indeed, interventions developed to prevent youths from initiating smoking often considered the need to involve parents [14,18–20]. For instance, a Cochrane review and meta-analysis of family-based interventions for smoking prevention among children and adolescents concluded that family-based interventions are effective to reduce the overall risk of smoking onset by 24% among intervention group compared to control group [18]. Systematic reviews were also conducted on parent-based interventions to reduce substance use among adolescents, presenting both success and failure of individual interventions to carry out intended outcomes [14,19,20]. However, these reviews did not conduct a meta-analysis of the included studies. Meta-analyses are indeed crucial to quantify the effects of interventions, therefore contributing to a complete appraisal of their potential benefits.

Tobacco use, especially smoking is a major determinant of health inequality worldwide [21]. The global distribution of the prevalence of tobacco use differs according to countries' socio-economic indices. For example, around 80% of the global tobacco users live in the low- and lower middle-income countries [2]. In contrast, most of the smoking prevention and cessation interventions are implemented in high-income countries [18,22]. Even in high-income countries, smoking prevalence varies according to socio-economic status, and adolescents make no exception [23,24]. The socio-economic gradient of smoking may imply that the effects of interventions might differ across socio-economic groups. This phenomenon has been labelled as “equity harm” associated with public health interventions where privileged groups benefit more from the interventions than underprivileged groups, which is mainly due to poor adherence to the intervention by participants from low socio-economic backgrounds [25,26]. Therefore, it would be important to understand if these interventions showed comparable effectiveness across socioeconomic groups, which is referred to as equity outcome in this review. To our knowledge, equity outcomes of smoking prevention and cessation interventions for young people, including parental involvement, have not been reviewed so far.

Therefore, the aim of this study is to conduct a systematic review and meta-analysis to determine the effectiveness of smoking prevention and cessation interventions for children and adolescents (≤ 18 -year-old) based on active parental involvement as well as potential threat to health equity arising from those interventions.

Methods

Design of the review

This review was designed based on 3 questions following the Population, Intervention, Comparator, and Outcome (PICO) frameworks (Table 1) [27]. The study protocol was registered in

PROSPERO with ID: CRD42023425348. Preferred Reporting Items for Systematic Review and Meta-Analysis was used as the reporting guideline [28].

Data sources and search strategy

A systematic literature search was conducted in Medline, Web of Science, PsycINFO, Embase, PubMed, [ClinicalTrials.gov](https://clinicaltrials.gov/), EU Clinical Trials Registrar, and WHO international clinical trials registry platform to identify studies that fulfilled the PICO criteria. Furthermore, a manual search was performed in Google Scholar using keywords and based on previously published systematic reviews on relevant topics. The search strategy is provided in the [Supplementary File](#). The literature search was conducted between 24 November 2022 and 27 November 2023.

Inclusion and exclusion criteria

For title and abstract screening, 5 criteria were used as follows: (1) experimental or quasi-experimental design; (2) smoking initiation and/or smoking cessation or reduction as outcome variables; (3) children and/or adolescent samples; (4) reporting original empirical data (i.e., not commentaries, debate articles, congress abstracts); and (5) published in English in peer-reviewed journals. For full-text screening, the following criteria were added: (1) the interventions should encompass an active parental involvement, where parents play an instrumental role in intervention delivery (i.e., not just being informed about the intervention or providing consent to allow children and adolescent to participate), ranging from driving teen to the intervention or receiving intensive skill training themselves; (2) the study samples should consist of general populations, i.e., not clinical samples; (3) the behavioural outcomes should be reported for smoked tobacco (e.g. cigarette, water pipe) separately from other tobacco or e-cigarette; and (4) the outcome should be measured at least 6 months after baseline assessment or postintervention initiation. For PICO 1, the following additional inclusion criteria were considered: (1) the intervention should aim to prevent smoking initiation; and (2) the behavioural outcomes should be defined as smoking initiation (lifetime smoking) among never smokers. Interventions were excluded if the outcome measures did not allow the separation between ever and current smoking. For PICO 2, additional inclusion criteria were as follows: (1) interventions aiming for smoking cessation; and (2) outcome defined as any measure of cessation of smoking among baseline current smokers. Finally, for PICO 3, an additional criterion required that the intervention effects were reported according to parental socio-economic status, as defined by income, education, or any validated measure of social class.

Selection process

The Karolinska Institutet's online library was used to conduct the literature searches. The citations were imported to Covidence (<https://app.covidence.org/>) where duplicates were removed. Thereafter, titles and abstracts were screened, followed by full-text screening. In both steps, records were independently screened by 2 reviewers (M.A.R. and B.Q.) and the conflicts were resolved by another reviewer (M.R.G.). Finally, data extraction and synthesis from the included articles were performed by one reviewer (M.A.R.) and crosschecked by another reviewer (M.R.G.), with conflicts resolved through mutual discussions.

Table 1
PICO frameworks for the review questions

Question	Population (P)	Intervention (I)	Comparator (C)	Outcome (O)
1	Children and adolescents never smokers	Smoking prevention interventions actively involving parents	No intervention or interventions without parents' involvement	Uptake of smoking
2	Children and adolescents current smokers	Smoking cessation interventions actively involving parents	No intervention or interventions without parents' involvement	Smoking cessation or reduction
3	Children and adolescents never or current smokers	Smoking prevention and/or cessation interventions involving parents	No intervention or interventions without parents' involvement	Difference in intervention's effect according to socio-economic status of the parents

PICO = Population, Intervention, Comparator, and Outcome.

Data extraction and descriptive analysis

The following information was aimed for extraction: identity of the records, intervention settings, scope of the intervention (prevention and/or cessation), tobacco type (cigarette and/or other smoked tobacco), age of the included sample, baseline prevalence of lifetime smoking, intervention components, description of parental involvement, study design, sample size, follow-up time, outcome measures, intervention effects (relative risks (RRs) with confidence intervals), and intervention effects by parental socio-economic status. Study characteristics and intervention effects are presented in summary tables. A two-way scatter plot with the best-fitted line was sketched to observe the relationship between age characteristics of the participants and lifetime smoking prevalence at baseline reported in individual studies.

Data synthesis and meta-analysis

The meta-analysis of preventive interventions (PICO 1) aimed to estimate the pooled RR of smoking initiation among never smokers who received interventions involving parents compared with never smokers in control groups receiving interventions without active parental involvement or no intervention. The selection of the studies to be included in the meta-analysis was done with the following criteria: a) for each intervention, only the study with the longest follow-up period was included, to avoid giving an unbalanced weight to interventions with outcomes reported on several follow-up points, and b) studies were excluded if they did not report the quantitative information needed to derive estimates of interventions' effects. In these cases, an attempt was made to collect the information by request to the corresponding authors.

If a study reported an evaluation of multiple interventions or employed more than one comparison group, we included in the meta-analysis the groups with active parental involvement and the comparison groups with no parental involvement or no intervention. Both published and unpublished records, including the supplementary files, were considered for data synthesis. Relative measures of effect, broadly defined as RR, included risk ratios, odds ratios, and hazard ratios [29]. The width of the 95% confidence interval was used to derive an estimate of the variance of the RR on the natural log scale. Unadjusted estimates were used from randomized controlled trials, while from quasi-experimental studies the reported adjusted effect estimate was used. If such estimates were not reported in the published articles, unadjusted RRs were derived from the available data. When the comparison was reported as control versus intervention group, the estimates were reversed before meta-analysis. Interventions with joint effects on all types of smoked tobacco were included in the meta-analysis. If this information was not

available, only the effect estimate for cigarette smoking was considered. Three smoking prevention studies were excluded from the meta-analysis because data synthesis was not possible. Meta-analysis was not carried out for studies addressing questions in PICO 2 and PICO 3 because of the low number of studies addressing the corresponding outcomes.

Meta-analysis was conducted using Stata software (version 17 StataCorp, College Station, Texas). A random-effect model was used for meta-analysis considering heterogeneity in study designs (individual randomised controlled trial, cluster randomised controlled trial, and quasi-experimental) and intervention content (a wide range of parental involvement), therefore assuming study-specific true effects. We conducted a subgroup analysis by the duration of follow-up (≤ 1 year and > 1 year). A leave-one-out meta-analysis was conducted to estimate how omitting any intervention would impact on the pooled effect estimates [30]. Meta funnel plot was used to explore small study effects [31]. A (posterior) predicted distribution of intervention effects in potential future studies was derived based on the estimated summary and variance of the study-specific effects [32].

Assessment of risk of bias and strength of evidence

The risk of bias of the included studies was assessed independently by one reviewer (M.A.R.) and crosschecked by another reviewer (M.R.G.), using the Cochrane risk of bias assessment tools (<https://www.riskofbias.info/>). Conflicts were resolved through mutual discussion. The risk of bias for experimental studies was evaluated for 5 domains as follows: randomization, deviation from the intended intervention, missing outcome data, measurement of outcome, and selection of the reported results. For quasi-experimental studies, bias was assessed for the following domains as follows: confounding, participant selection, classifications of interventions, deviation from intended interventions, missing data, outcome measurement, and selection of the reported results. The inter-rater reliability (Cohen's kappa) for title and abstract screening and full-text screening were retrieved from Covidence.

For each outcome, the strength of evidence was assessed by the Grading of Recommendations Assessment, Development, and Evaluation (GRADE), with domains related to risk of bias, inconsistency, indirectness, imprecision, publication bias, and overall effect [33]. The strength of evidence was assessed using the GRADEpro (<https://www.gradepro.org/>) software.

Results

Selection of the studies

A total of 6,748 records were retrieved for screening. After removing 2,651 duplicates, 4,097 records were screened for titles

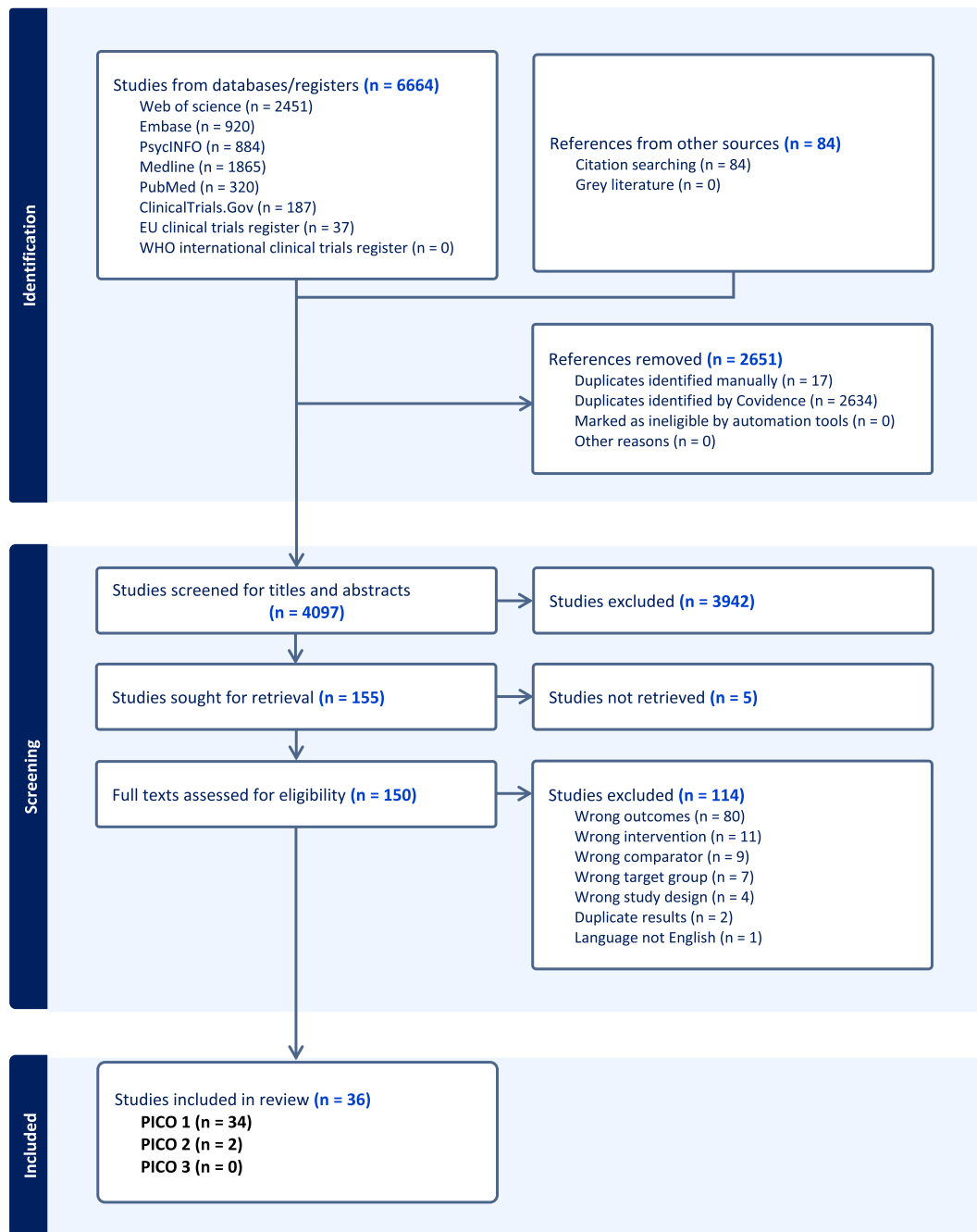


Figure 1. Flow diagram of the study selection (n = 36 studies from 29 unique interventions).

and abstracts and 150 records for full texts. Finally, 36 published studies were included in the review, reporting on 29 unique interventions. Of these interventions, 27 concerned smoking prevention (PICO 1) and 2 concerned smoking cessation (PICO 2). None of the included articles reported results on intervention effects according to parental socio-economic status (PICO 3). [Figure 1](#) displays the selection process. The inter-rater reliability (Cohen's kappa) for titles and abstracts screening and full-texts screening were 0.57 and 0.86, respectively, indicating a high level of consensus.

Characteristics of the studies

Twenty-seven out of 29 interventions were evaluated through experimental designs, while 2 were evaluated through quasi-experimental designs. The included studies were published over a wide range of years (1995–2022), with almost one-fourth (27.8%) published in the last 10 years. Most studies were conducted in the United States (72.2%) and Europe (22.2%), with only one study being conducted in Australia [34] and one in India [35]. Most of the interventions targeted early adolescents

Table 2

Setting, design, and participants' characteristics of the included studies (n = 36, 29 unique interventions)

Study reference#	Country	Setting	Design	Sample size ^a	Age (years) ^b
Intervention focus: smoking prevention					
[36]	Germany	School	RCT	292	12
[37]	USA	Home	RCT	1,316	14
[38]	Sweden	School	cRCT	1,776	13
[39]	USA	School	cRCT	1,034 (cohort 1), 1,244 (cohort 2)	10, 12
[40]	USA	Home and health care	RCT	4,026	11
[41]	Several ^c	Multicenter	cRCT	19,034	13
[42]	USA	School	cRCT	7,827	8
[43]	USA	School	cRCT	4,689	12
[44]	Poland	Community and home	cRCT	614	12
[45]	USA	School	RCT	1,386	12
[46]	USA	Home	RCT	331	14
[47]	USA	School	Quasi-experimental	643	11
[48–50]	USA	Community	cRCT	4,407	10
[51]	Netherlands	Home	cRCT	1,478	10
[34]	Australia	School	cRCT	3,115	13
[52]	USA	Home	RCT	873	8
[53]	Germany	School	cRCT	2,801	13
[54]	USA	Home	RCT	266	13
[35]	India	School	cRCT	4,776	12
[55]	USA	Community	cRCT	167	11
[56]	Netherlands	Home	RCT	512	11
[57–59]	USA	School	cRCT	667	12
[60,61]	USA	School	cRCT	1,677	12
[62,63]	USA	School	cRCT	11,960	11
[64]	UK	School	Quasi-experimental	2,137	11–13
[65,66]	USA	School	RCT	563	11
[67]	USA	School	cRCT	358	12
Intervention focus: smoking cessation					
[68]	USA	Home	RCT	85 ^d	13
[69]	USA	Hospital	RCT	162 ^d	14–18

cRCT = Cluster Randomised Controlled Trial; RCT = Randomised Controlled Trial; UK = United Kingdom; USA = United States of America.

^a Participants at baseline.^b Presented as mean, range, or equivalent to academic grade/class. Fractions were rounded up to the nearest integers.^c Several countries included Denmark, Finland, Netherlands, Portugal, Spain, and UK.^d Tobacco users at baseline.

(10–13 years) (Table 2). The baseline prevalence of lifetime smoking among participants, reported in 16 interventions, showed an increasing trend from childhood to adolescence with high variation (2.4%–37.7%) between studies around the age of 12 (Supplementary File).

School was found to be the most common (55.2%) intervention setting, followed by home (24.3%). The rest of the interventions (20.7%) were delivered either at hospital or community settings. The most frequent types of parental involvement were general parenting support and child-parent communication (17 interventions), interventions fostering parental-specific antitobacco messages and behaviors (16 interventions), and support for parental smoking cessation (2 interventions) or combinations of these (Supplementary File).

Interventions effects

Smoking prevention interventions. The effects of 27 interventions aiming at preventing youth smoking are shown in the Supplementary File. The meta-analysis of 24 interventions resulted in a summary estimate of RR of smoking initiation among participants receiving interventions with parental components compared with control groups of 0.84 (95% CI: 0.76, 0.94), indicating a 16% lower risk of smoking initiation. The summary estimate of RR of the interventions with ≤1 year of follow-up and >1 year of follow-up were 0.69 (95% CI: 0.47, 1.02)

and 0.85 (95% CI: 0.76, 0.95), respectively (Figure 2). The overall degree of heterogeneity across interventions included in meta-analysis was high ($I^2 = 68.21\%$).

The 95% prediction interval for a future intervention effect was 0.57–1.25 (Figure 3). Leave-one-out meta-analysis showed minor changes of the estimated pooled effect (Supplementary File).

Smoking cessation interventions. Only 2 studies on smoking cessation based on parental interventions fulfilled the inclusion criteria (Table 2). Of these studies, one reported a 42% higher rate of smoking cessation in the intervention group compared with control [68]. In the other study, intervention and control groups had different rates of smoking cessation at follow-up (4.9% in intervention and 2.8% in control); however, this difference was not statistically significant [69] (Supplementary File).

Health equity effects of the interventions. None of the included studies reported intervention effects separately by parental socio-economic status or sufficient information to derive separate estimates. Therefore, it was not possible to evaluate the equity aspects of the included interventions.

Risk of bias and strength of evidence

All the included studies were judged to suffer from high or serious risk of bias, which was primarily due to the probability of

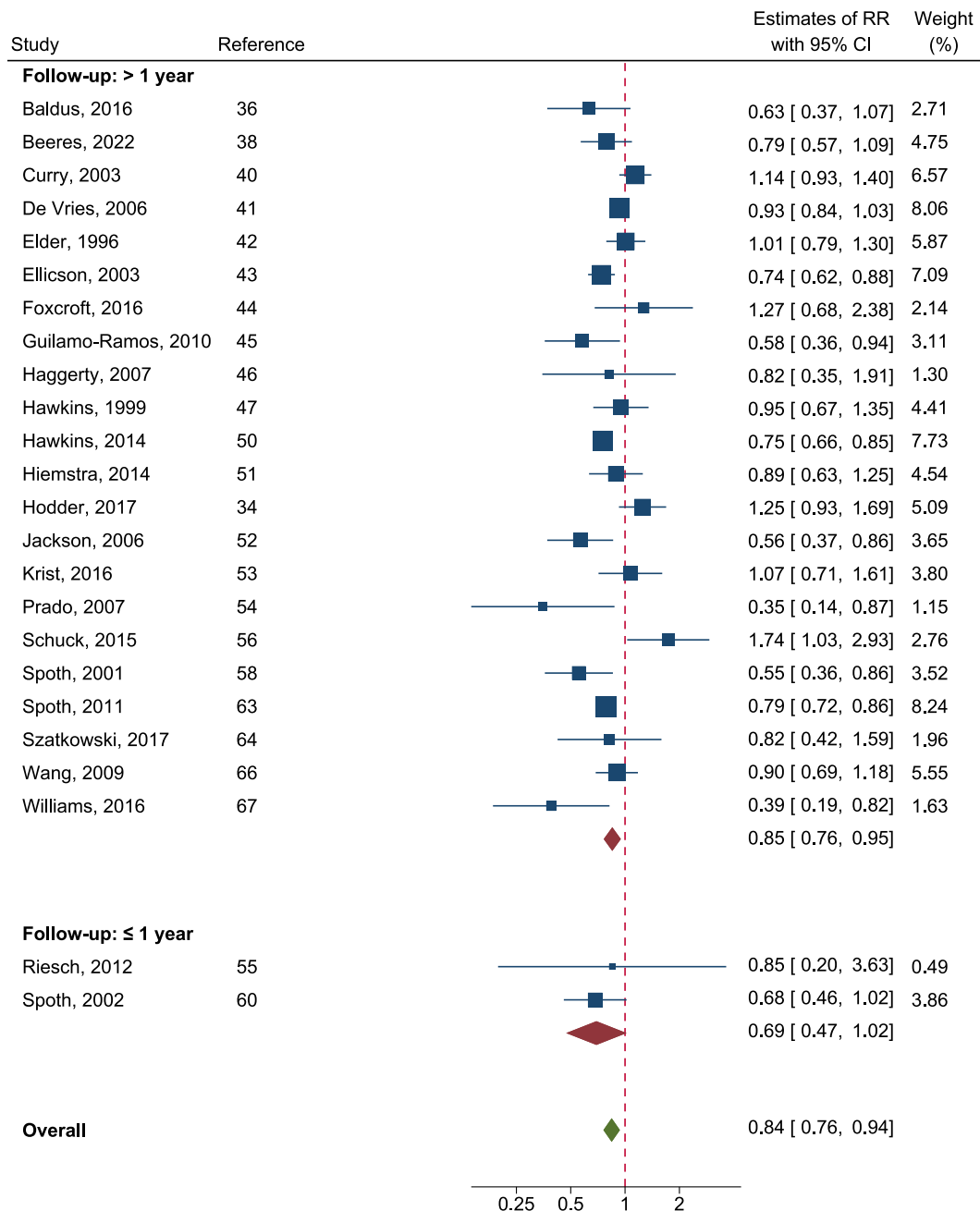


Figure 2. Pooled relative risk of smoking initiation among participants receiving interventions involving parents versus participants in control group. A meta-analysis of 24 studies.

unreliable outcome measures from self-reports ([Supplementary File](#)). Additionally, of the experimental studies, 18.5% ($n = 5$) had high risk for deviation from intervention [41,45,57,67,68] and 14.8% ($n = 4$) had high risk due to missing outcome [42,53,54,56], while none carried high risk for selection of reported result. Furthermore, more than half (51.9%, $n = 14$) and almost one-third (29.6%, $n = 8$) of the studies had some concerns of bias, respectively, for deviation from interventions and selection of reported results. Both quasi-experimental studies had serious risk of bias for confounding and outcome measurement. The overall strength of evidence provided by the meta-analysis was graded as moderate following the GRADE

criteria. The funnel plot of the studies included in the meta-analysis showed a slight over-representation of smaller studies among those showing a beneficial effect of the interventions, indicating a possible publication bias ([Figure 4](#)).

Discussion

Potential of preventive interventions involving parents

In this study, there was a moderate degree of evidence that preventive interventions, including active parental involvement,

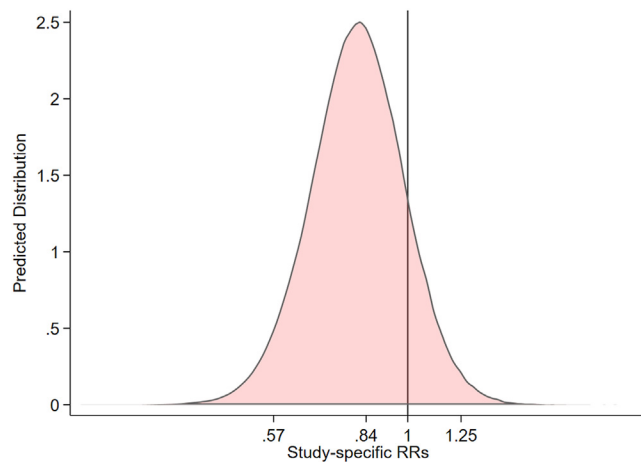


Figure 3. Predicted distribution of future intervention effects.

may delay the onset of smoking among children and adolescents, who are at high risk of smoking initiation. The scope of the parental involvement in these interventions fell into quite well-defined categories broadly matching the main determinants of parental influences on adolescent smoking ([Supplementary File](#)). However, it was not possible to analyze the effect of different types of parental interventions, because these were often embedded in different multicomponent intensive programs. For the same reason, it would be premature to attribute the whole estimated preventive effect solely to parental involvement. However, it can be noted that in 3 studies where the comparison group received a similarly intensive attention as the intervention group, but without parental involvement [35,37,39], the results clearly favoured the role of parental involvement as intervention condition.

The few studies with short-term follow-ups (≤ 1 year) presented more homogenous estimates of effects and somewhat a stronger association with delayed initiation than in studies with longer follow-ups. This may indicate that the beneficial effects tend to fade with time, possibly because the motivation to abstain from smoking might be influenced by other social and environmental factors in the life course [10]. However, loss of

power due to selective attrition at long follow-up may also play a role, rendering the compared group more like each other in their behavioral trajectories.

Smoking cessation interventions

There was insufficient evidence on the effects of parental involvement in promoting smoking cessation among adolescents. Only 2 smoking cessation studies with parental active involvement as an intervention component met the inclusion criteria for this review, one of which presented a favorable effect [68]. The scant scientific evidence can be due to the fact that smoking cessation interventions are usually delivered in managed care where parental involvement may prove difficult [70,71]. Also, the assessment of cessation outcomes requires the presence of a regular and established smoking behavior, which in its turn conveys a high risk for dependence, rendering behavioral modifications more responsive to psychopharmacological treatments than to social influences [72]. In fact, smoking in adolescence can hold for a long time very changeable patterns, with frequent spontaneous remissions and relapses [73], which may make it arduous to assess the stability of intervention effects at early stages.

Equity aspect of the interventions

None of the included studies reported the intervention effects separately in subgroups of socio-economic status of the family. Indicators of socio-economic circumstances were sometimes used to control for potential confounding [38,44] and/or as outcome predictors on their own [52] but not to elucidate potential differences likely to affect health equity outcomes. One study reported that the risk of smoking initiation was higher among adolescents of parents with high school education or lower than among adolescents whose parents had at least some college education [52]. This supports the notion that adolescents in lower socio-economic groups are a very important target for universal prevention. The reasons why low socio-economic groups may benefit less from preventive interventions are a matter of speculation, since we could not find studies that explored this pattern. Poor parental adherence to the intervention due to socio-economic stressors and unclear priorities or expectations of benefits may be an explanation [26]. Also, adolescents in the contexts of social disadvantage may be exposed to social, behavioural, and environmental factors that reinforce smoking, therefore counteracting preventive messages [74]. We suggest that the assessment of potential undesirable inequity impact should be a priority in future studies.

Novelty of current study

Only a few systematic reviews were conducted on smoking prevention and cessation interventions for children and adolescents, albeit not exclusively focusing on parental active involvement [14,19,20]. One review demonstrated that smoking-specific parenting, e.g., complete smoking house rules, had an effect in the desired direction on smoking onset among youth [14]. Other reviews highlighted the importance of involving parents in interventions targeting substance use reduction among adolescents [19,20]. However, in the current study, we quantified the overall effect of very specific interventions, those including parental active involvement, in delaying smoking onset among

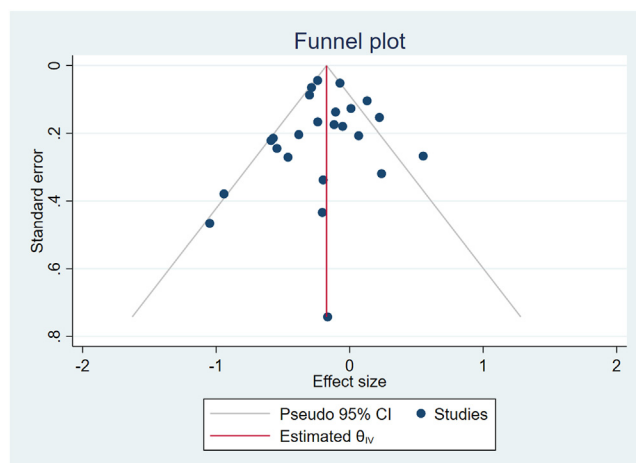


Figure 4. Funnel plot of the studies included in the meta-analysis.

children and adolescents. In this respect, the results of this review strengthen the conclusions of previous reviews that emphasized the importance of parental involvement in a more general way. Although, the scope of this review partially overlaps with that of a Cochrane review [18], we believe that this study contributes to the advancement of knowledge in this field in multiple ways. First, the current study included interventions with parents' involvement delivered in any setting, e.g., school and community, and not limited to households only. Second, we adopted a very strict measure for prevention outcomes, i.e., initiation of smoking or lifetime smoking. Third, the current meta-analysis presents updated evidence encompassing 24 relatively recent interventions of which 10 were evaluated after 2015. Fourth, we focused on elucidating 2 new outcomes apart from smoking initiation, i.e., cessation and equity.

Sources of heterogeneity

Despite the average favorable effect of parental interventions on juvenile smoking onset, quite a few studies with inconclusive or opposite effects could provide hints on the reasons for the heterogeneity of the effect estimates. In studies adopting interventions that were initially evaluated in other contexts, inadequate adaptation to the new context may be a reason for not achieving the desired impact [34,44], since cultural factors (e.g., role of the parents in a given society) may modify the intervention effects. For instance, the Strengthening Family Program (10–14), designed for the USA context, was not associated with favourable outcomes in Poland [44]. A short-term follow-up was sometimes indicated as a reason [34,44,53]. Along with excessive emphasis of the intervention on individual level factors rather than on family and environmental factors [34], low intensity of the interventions [53] and selection of appropriate target groups should also be considered. Factors related to the evaluation methods may refer to the use of pragmatic experiments [34], possible contamination [34], and low validity of the outcome assessment [34].

Additional sources of heterogeneity could be due to the settings, the population prevalence of smoking, participant's age, core intervention components, nature of the comparison condition, types of parental involvement, duration of follow-up, methods of analysis, and variables adjusted for in the analysis.

Quality of the evidence

The high risk of bias common to all included studies concerned the potential reporting bias in the outcome assessment, given the lack of blindness, with consequent awareness of participants and often also of assessors about the interventions' nature and expectations. This bias, if determined by social desirability, would generally favor the intervention group. Unfortunately, biochemical validation may not be a viable remedy for unreliable self-reports due to their insensitivity to distant smoking episodes [72], as is the case in youth's lifetime smoking. Furthermore, concerns remain about the feasibility and ethics of biochemical verification of smoking onset.

The moderate strength of the evidence was mainly due to the serious risk of bias, inconsistency of the results, and publication bias of the studies. Apart from the validated quality assessment measures, it was noticed that participants' age, baseline prevalence, and sample size were not clearly reported in some studies.

Strengths, limitations, and future scope

This study adds to the scientific literature by extending the focus and the knowledge of previous reviews, specifically a stringent focus on parents rather than on family at large, including both smoking prevention and cessation interventions, and including more recent evidence. A careful search for both published and unpublished records was done, with the scope of minimizing publication bias; the detailed a priori definition of inclusion and exclusion criteria guided by the advocated framework for research questions [27], the processes of screening and selection of the studies, and data extraction that were carried out with strict adherence to established guidelines [28] and with high level of consensus among reviewers.

Notwithstanding, we need to acknowledge some limitations. Studies published in languages other than English were not included, with potential under-representation of studies from low-income countries. The nature of the interventions and, above all, the comparison groups make it difficult to disentangle the effects of parent involvement from those of other components. Also, the nature of parental involvement varied considerably between interventions, contributing to the high heterogeneity of the effect estimates. Three studies of smoking prevention interventions could not be included in the meta-analysis because data synthesis was not possible. Therefore, uncertainty remains whether the pooled effect size might have been biased by these exclusions. However, the predicted distribution of simulated replications of similar studies makes such radical change rather unlikely (Figure 3). Publication bias, both a methodologic and an ethical concern, could not be excluded [75].

The findings suggest that overall, smoking prevention interventions with active parental involvement seemed to be moderately effective in delaying the onset of smoking among children and adolescents. However, assessing the equity aspects of smoking prevention and cessation interventions with parental involvements should be a priority for future evaluations. Parental involvement could be considered in smoking cessation interventions, and their effectiveness should be assessed. Also, the effectiveness of different approaches to parental involvement and the moderating effect of specific contexts should be addressed prior to policy recommendations. In future studies, the effectiveness of different strategies of parental involvement should be addressed, which will be important for subsequent judgments of the costs and benefits of their scaled implementation.

Conclusion

Smoking prevention interventions that actively involve parents appear to be effective in delaying the onset of smoking among children and adolescents. Given the moderate strength of evidence and high heterogeneity across the studies, this study emphasizes quality outcome assessment and a standardized approach to involving parents in future interventions. The evidence on the effect of interventions involving parents on smoking cessation among children and adolescents is inconclusive, and additional studies are required in this field. Health equity is an unexplored and very important aspect in the evaluation of smoking prevention and cessation interventions involving parents of children and adolescents.

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Ethics Approval

Ethics approval was not sought because data was not collected from any human participants.

Data Availability Statement

Additional information could be accessed through the corresponding author upon reasonable request.

Supplementary Data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jadohealth.2024.10.032>.

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